



CPSC 100





CPSC 100

Computational Thinking

Syllabus + Introduction

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Land Acknowledgement

I acknowledge that I am
presenting this session today on
the **traditional, ancestral, and**
unceded territory of the
xwməθkwəy̓əm
(Musqueam) peoples.



Plan for Today

- About Me
- About You
- Syllabus
- Computation Thinking



Firas Moosvi

Lecturer

[University of British Columbia](#)



Biography

I am a Lecturer in the Computer Science department at the University of British Columbia. Though I mainly teach computer science now, I am a multidisciplinary educator with a PhD in Physics and is also interested in data science and education in general. I strongly believe in computational literacy for all and aims to make STEM courses accessible through Active Learning techniques and open education resources. My two main research umbrellas are the scholarship of teaching and learning (SoTL), and Learning Analytics. I am looking at how learning analytics data can provide insight to surface and ultimately reduce inequities in STEM programs. I am also heavily invested in promoting and implementing alternative grading systems in large classes, at scale. I am always happy to collaborate on teaching and learning projects, [drop me a note here!](#)

Interests

- Scholarship of teaching and learning
- Authentic assessments
- Alternative grading paradigms
- Learning analytics
- Data visualization and science communication

Education

-  PhD in Medical Physics, 2019
University of British Columbia
-  MSc in Medical Biophysics, 2012
University of Toronto
-  BSc in Biophysics, 2009
University of British Columbia

Research Interests



Learning Technologies

Use of learning technologies to enhance teaching and learning.



Active Learning

A learning method that de-emphasizes didactic teaching and actively engages students with material via problem solving, case studies, role plays and other methods.



Learning Analytics

Extracting trends from learner data using analytical tools to improve learning.



Equity in STEM

Developing and implementing methods of inclusive teaching to reduce systemic inequities in STEM education.



Visualizations

Representing data using effective graphs, plots, and other special visualizations.



Alternative Grading

Challenging the systems and structures associated with traditional grading in higher education.

Learning Goals

What are Learning Goals?

Learning Goals

A learning goal is designed to tell you what information you should learn about a topic

- At the beginning of the topic, the learning goal may not even make any sense
- By the end of a topic, it should help you know what to study

Learning Goals for the course

Learning Goals

At the highest level, this courses has three main goals.

Students who complete this course will be able to

- explain foundational computing concepts,
- explore applications built on these foundations,
- describe the implications of the applications.

**How are you
going to learn?**



How to learn?

The best way to learn is by doing

Weekly Labs - in labs you will

- Get hands-on practice with a computer
- Work on projects

“Lectures” will involve

- Doing exercises
- Discussing the readings (in small and large groups)
- Hearing concepts explained
- Answering clicker Questions



What does this mean for you?

You will need to read and review slides for the next class before coming to class.

Sometimes, there will be assigned pre-class readings or videos.

You need to come to class prepared to *do* things:

- Bring a web-enabled device (tablet, laptop, even a phone will do)
- Follow-along the slides posted on the course website
- Be prepared to be **ACTIVE!**

About this course



About this course

CPSC 100 - Computational Thinking [3]

- Monday / Wednesday / Fridays, 3-4 PM in ESB 1012
- “Lectures” will involve:
 - In class activities on **PrairieLearn**
 - Clickers using **Agora**
 - Discussions in person
- Lectures will not be recorded



What this course includes



Course Schedule

This is the tentative plan for CPSC 100 this term. This is subject to change

Week	Week of	Topic
1	Jan 5	Introductions
2	Jan 12	Algorithms I
3	Jan 19	Algorithms II
4	Jan 26	Data Representation
5	Feb 2	Programming I
6	Feb 9	Programming II
7	Feb 16	Reading week
8	Feb 23	Internet and HCI
9	Mar 2	Artificial Intelligence I
10	Mar 9	Artificial Intelligence II
11	Mar 16	Catch-up week
12	Mar 23	Data Mining
13	Mar 30	Data Visualization
14	Apr 6	Final Projects



What this course is NOT

1) A typical Computer Science course

- You will do more reading/writing than coding

2) A coding-based course

- You will not really be programming (only block-based)

3) An “easy A”

- Course is set up reward students who do the work!

This is **NOT** a
"GPA booster class"





Course Logistics

Course website

firas.moosvi.com/courses/cpsc100/2025W2

Communication

Ed Discussion (join via link on Canvas)

It is your responsibility to check Ed for announcements

At least once a day!

Class Etiquette



Course Etiquette

Be professional, respectful and thoughtful

Attend class on time and prepared

Don't distract others

Please stay home if you are not feeling well

Review posted course material

Contact classmates to catch up



Professionalism

A high level of professionalism is expected
Respectful communication in class, emails, etc.

A typical Ed Discussion Post:

Tag the post correctly (Lab, LL, Test, Logistics, etc.)

A greeting, a clear message, a closing

Using ChatGPT for posts is NOT recommended

Blatant violations of this will be returned for revision



My Expectations of you

Reach out if you need help

Academic, mental health, etc.

Be honest

Academic integrity, always ask if you're unsure

Be a considerate team member

Attend (on-time) and participate in meetings

Do your share of the job, well and on time.

Syllabus Review

slido



How are you feeling about this course?

① Click **Present with Slido** or install our [Chrome extension](#) to activate this poll while presenting.

Computational Thinking

slido



What do you think Computational Thinking is?

① Click **Present with Slido** or install our [Chrome extension](#) to activate this poll while presenting.

What is Computational Thinking?

“Computational thinking is the **thought processes** involved in formulating problems and their solutions so that the solutions are in a form that can be effectively carried out by an information-processing agent” [Cuny, Snyder, Wing 10].



DECOMPOSITION

Breaking down problems into smaller, easier parts.



PATTERN RECOGNITION

Using patterns in information to solve problems.



ABSTRACTION

Finding information that is useful and taking away any information that is unhelpful.



MODELLING AND SIMULATION

Trying out different solutions or tracing the path of information to solve problems.



ALGORITHMS

Creating a set of instructions for solving a problem or completing a task.



EVALUATION

Assessing a solution to a problem and using that information again on new problems.

ACTIVITY (Groups of 5)

Discuss in your groups the following question:

Which of the 6 computational thinking skills do you think **are the most important (rank them)?**

Which do you think are the **most intuitive?**



Wrap up



Wrap Up

- Labs start next week
- Complete ASAP:
 - Read the Syllabus
 - Syllabus Quiz (Test 0) - not for real marks
- First test (Test 1) is next week!
 - More details on Wednesday